Abstract

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This paper presents Deterministic bounded AutoResearch fo r a small MNIST neural-network task, a public template exemplar that turns an AutoResearch loop into ordinary reproducible research infrastructure. The case study is intentionally small but concrete: 2000 training and 500 test images from MNIST handwritten digit database are evaluated by the bounded sm all MNIST neural-network classification loop. The run evaluates 4 of 5 proposed candidates, including Tiny patch-atte ntion classifier, selects exp-mlp-tanh-64 (MLP, 50890 parameters), and improves test_accuracy from 82.6% to 89.4% (6.8% absolute change). The validated diagnostic layer reports macro F1 89.4%, bootstrap accuracy interval 86.4% to 92.0%, Brier score 0.161, negative log

likelihood 0.361, top-2 accuracy 95.6%, and exact McNemar p-value 0.000. The same pipeline writes proposal, candidate, run, review, benchmark, evidence, figure, confusion-matrix, statistical-summary, probability-quality, and security-integrity artifacts from declared output contracts; uses 0 LLM calls at USD 0.00 cost; and records 7 configured stages, 6 supported local-artifact claims, and 78 required artifacts. The local security attestation status is passed, with 0 checksum mismatch(es). The final readiness status is passed, with review gates deferred to a human rather than self-approved by the generated run.